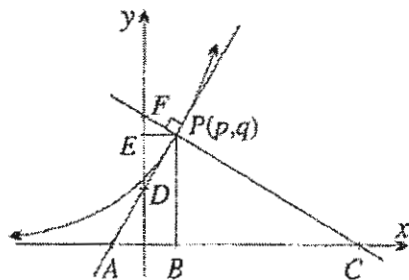


Sketching Exponential Functions

1. Use your knowledge of transformations to help sketch the graphs of the given functions:
 - a. $y = e^{x-1}$
 - b. $y = e^{-x}$
 - c. $y = -e^x$
 - d. $y = e^{x-2}$
 - e. $y = 1 - e^x$
 - f. $y = e^{-x} - 1$
 - g. $y = e^{\frac{1}{2}x}$
 - h. $y = e^{-|x|}$
2. a. Find the equation of the tangent to $y = e^x$ at its y-intercept.
 - b. Show that the tangent to $y = x - e^x$ at $x = 1$ passes through the origin
3. a. Find the equation of the normal to $y = e^{-x}$ at the point $P(-1, e)$.
 - b. Find the x and y intercepts of the normal.
 - c. Find the area of the triangle whose vertices lie at the intercepts and the origin.
4. a. Find the first and second derivatives for the curve $y = x - e^x$.
 - b. Deduce that the curve is concave down for all values of x .
 - c. What is its maximum value?
 - d. Sketch the curve and write down its range.
5. Consider the curve $y = xe^x$
 - a. Show that $y' = (1 + x)e^x$ and $y'' = (2 + x)e^x$.
 - b. Show that there is one stationary point and determine its nature.
 - c. Find the coordinates of the lone point of inflexion.
 - d. Examine the behaviour of y as $x \rightarrow -\infty$.
 - e. Hence sketch the curve, and then write down its range.
 - f. Hence also sketch $y = -xe^{-x}$ by recognising the simple transformation.
6. a. Given that $y = e^{-\frac{1}{2}x^2}$, find y' and y''
 - b. Show that this curve has a maximum turning point at its y-intercept and has two points of inflexion.
 - c. Examine the behaviour of y as $x \rightarrow -\infty$ and $x \rightarrow \infty$.
 - d. Sketch the graph and write down its range.
7. a. Given that $y = (1 - x)e^x$, find y' and y'' .
 - b. Show that this curve has a maximum turning point at its y-intercept, and an inflexion point at $(-1, 2e^{-1})$.
 - c. Sketch the graph and write down its range.
8. Find the x -intercept of the tangent to $y = (1 - x)e^x$ at $x = -1$.
9. [Another characterisation of $y = e^x$ – it is only exponential function whose gradient at its y-intercept is 1]
 - a. Prove that $y = a^x$ has derivative $y = a^x \log a$
 - b. Prove that $y = a^x$ has gradient 1 at its y-intercept if and only if $a = e$.
 - c. Prove that $y = Aa^x$ has gradient 1 at its y-intercept if and only if $a = e^{\frac{1}{A}}$.
10. The line $y = mx$ is tangent to the curve $y = \frac{e^x}{x}$. Show that the point of contact is $A\left(2, \frac{1}{2}e^x\right)$ by showing that the gradient of OA is equal to the gradient of the tangent to the curve at A .
11. a. Find the equation of the tangent to $y = e^x$ at $x = t$.
 - b. Hence show that the x -intercept of this tangent is $t - 1$. Does this agree with your answers to question 2(d)?
12. a. Show that $y = e - x^2e^x$ has an x -intercept at $x = 1$.
 - b. Show that the curve has two turning points and classify them
 - c. Examine the behaviour as $x \rightarrow \infty$ and deduce that it also has two inflexion points.
 - d. Sketch the curve and write down its range.
13. a. Find the intercepts of $y = (1 + x)^2e^{-x}$.
 - b. Show that the curve has two turning points and classify them.
 - c. Examine the behaviour of y as $x \rightarrow \infty$ and hence deduce that it also has two inflexions.
 - d. Sketch the curve and write down its range.
14. a. Classify the stationary points of $y = xe^{-x^2}$.
 - b. Locate the three inflexion points, sketch the curve and write down its range.
15. a. Find the equation of the normal to $y = e^{-x^2}$ at the point where $x = t$.
 - b. Determine the x -intercept of the normal.
 - c. Hence find the values of t for which the normal passes through the origin.

16. On the graph of $y = e^x$ are drawn the tangent and normal to the curve at the point $P(p, q)$.

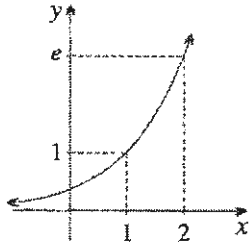


- Find the coordinates of each of the points A, B, C, D, E and F in terms of p and q .
 - Hence show:
 - $AB = 1$
 - $BC = q^2$
 - $DE = pq$
 - $EF = \frac{p}{q}$
 - What is the area of:
 - $\triangle ACP$?
 - $\triangle DFP$?
17. Find the x -coordinates of the stationary points of $y = xe^{-|x|}$ by considering positive and negative values of x separately.
18. Show that $y = (x^2 + 3x + 2)e^x$ has an inflexion point at one of its x -intercepts. Sketch the curve and label all important features. Do not find the y -coordinates of the stationary points.
19. a. What is the natural domain of $y = \frac{e^x}{x}$?
 b. Show that the curve has a local minimum at $(1, e)$ but no inflexion points.
 c. Sketch the curve and states its range.
20. a. What is the natural domain of $y = e^{\frac{1}{x}}$?
 b. Carefully determine the behaviour of y and y' as $x \rightarrow -\infty, x \rightarrow 0$ and $x \rightarrow \infty$.
 c. Deduce that there must be an inflexion point and find it
 d. Sketch the curve and give its range.
21. Follow the steps in the previous question in order to sketch the graph of $y = xe^{\frac{1}{x}}$.
22. If $x = e^{-4t}$, find the functions v and $\ddot{x}$.
- Explain why neither x nor v nor $\ddot{x}$ can ever change sign, and state their signs.
 - Where is the particle:
 - Initially,
 - Eventually?
 - What are the particle's velocity and acceleration
 - Initially,
 - Eventually?
23. A large stone is falling through a layer of mud, and its depth x metres below ground level at time t minutes is given by $x = 12 - 12e^{-0.5t}$.
- Find v and $\ddot{x}$ as functions of t , and sketch graphs of x, v and $\ddot{x}$.
 - In which direction is the stone:
 - Travelling,
 - Accelerating?
 - What happens to the position, velocity and acceleration of the particle as $t \rightarrow \infty$?
 - Find when the stone is halfway between the origin and its final position. Show that its speed is then half its initial speed, and its acceleration is half its initial acceleration.
 - How long, correct to the nearest minute, will it take for the stone to reach within 1 mm of its final position?
24. Two particles A and B are moving along a horizontal line, with their distances x_A and x_B to the right of the origin O at time t given by $x_A = 4te^{-t}$ and $x_B = -4t^2e^{-t}$. The particles are joined by a piece of elastic, whose midpoint M has position x_M at time t .
- Explain why $x_M = 2e^{-t}(t - t^2)$, find when M returns to the origin, and find its speed and direction at this time.
 - Find at what times M is furthest right and furthest left of O .
 - What happens to A, B and M eventually?
 - When are A and B furthest apart?

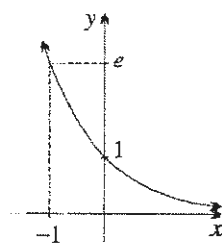
Answer

1.

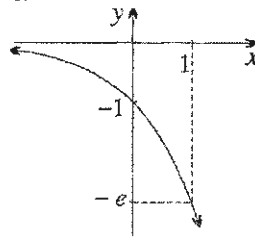
a.



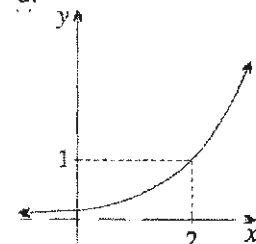
b.



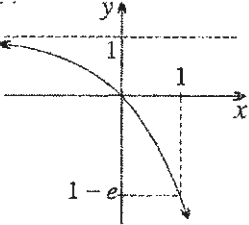
c.



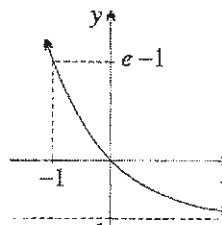
d.



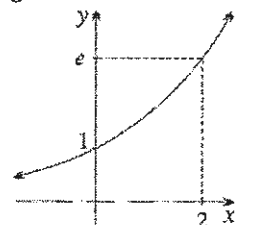
e.



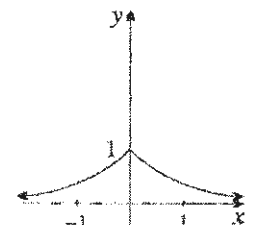
f.



g.



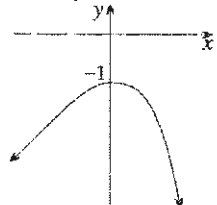
h.



2. a. $y = x + 1$; b. $y = (1 - e)x$

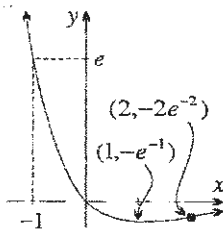
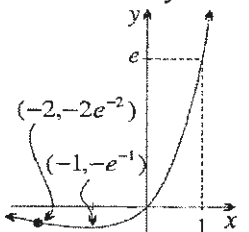
3. a. $x - ey + e^2 + 1 = 0$; b. $x = -e^2 - 1, y = e + e^{-1}$; c. $\frac{1}{2}(e^3 + 2e + e^{-1})$

4. a. $y' = 1 - e^x, y'' = -e^x$; d. $y \leq 1$

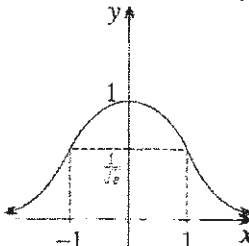


5.

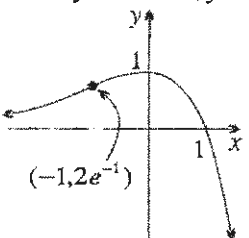
e. $y \geq -e^{-1}$



6. a. $y' = -xe^{-\frac{1}{2}x^2}, y'' = (x^2 - 1)e^{-\frac{1}{2}x^2}$; d. $0 < y \leq 1$



7. a. $y' = -xe^x, y'' = -(x + 1)e^x$; c. $y \leq 1$



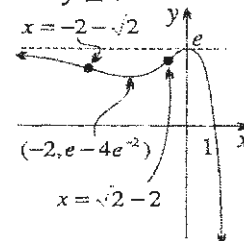
8. -3

9. b. the gradient of $y = a^x$ at $x = 0$ is $\log a$, which is 1 and only if $a = e$; c. the gradient of $y = Aa^x$ at $x = 0$ is $A \log a$, which is 1 if and only if $a = e^{\frac{1}{A}}$.

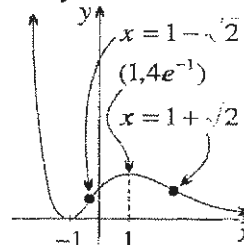
10. prove

11. a. $y = e^t(x - t + 1)$

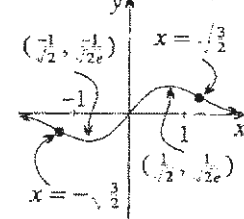
12. $y \leq e$



13. $y \geq 0$



14. $-\frac{1}{\sqrt{2}e} \leq y \leq \frac{1}{\sqrt{2}e}$



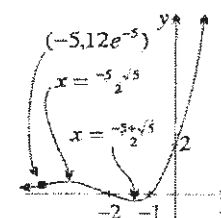
15. a. $x - 2te^{-t^2}y + t(2e^{-2t^2} - 1) = 0$; b. $t(1 - 2e^{-2t^2})$;

c. $-\sqrt{\frac{\log 2}{2}}, 0$ and $\sqrt{\frac{\log 2}{2}}$

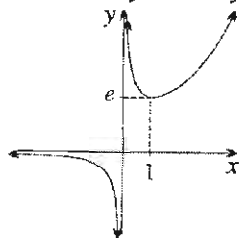
16. a. $A(p - 1, 0), B(p, 0), C(p + q^2, 0), D(0, (1 - p)q), E(0, q), F(0, q + \frac{p}{q})$; c. i. $\frac{q}{2}(q^2 + 1)$, ii. $\frac{p^2}{2}(q + \frac{1}{q})$

17. $x = 1$ or $x = -1$

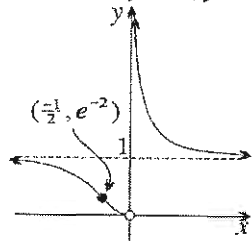
18.



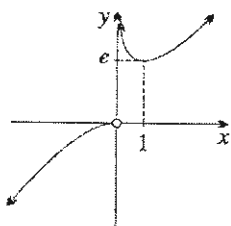
19. $x \neq 0, y < 0$ or $y \geq e$



20. $x \neq 0, y > 0, y \neq 1$

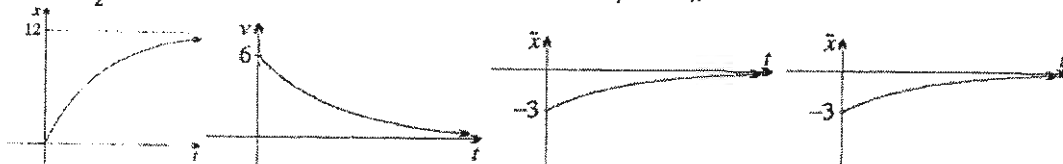


21.



22. $v = -4e^{-4t}, x = 16e^{-4t}$; a. e^{-4t} is positive for all t , so v is always negative, and $\ddot{x}$ is always positive; b. i. $x = 1$, ii. $x = 0$; c. i. $v = -4, \ddot{x} = 16$, ii. $v = 0, \ddot{x} = 0$

23. a. $x = 12 - 12e^{-0.5t}, v = 6e^{-0.5t}, \ddot{x} = -3e^{-0.5t}$ (graph see below); b. i. downwards (Downwards is positive here.) ii. upwards; c. the velocity and acceleration tend to zero, and the position tends to 12 metres below ground level; d. $x = 6$ when $e^{-0.5t} = \frac{1}{2}$, that is, $t = 2 \log 2$, the speed then is 3 m/min (half the initial speed of 6 m/min) and the acceleration is $-1\frac{1}{2} \text{ m/min}^2$ (half the initial acceleration of -3 m/min^2); e. 19 minutes



24. a. The displacement of M is the average of x_A and x_B , $v_M = -2e^{-t}(t^2 - 3t + 1)$, M returns to O after 1 second, when M is moving left at a speed of $\frac{2}{e}$; b. the particle is furthest right at $t = \frac{1}{2}(3 - \sqrt{5})$, and furthest left at $t = \frac{1}{2}(3 + \sqrt{5})$; c. They all move towards O ; d. $t = \frac{1}{2}(1 + \sqrt{5})$.